SCOPE: When Generated Legal Queries Help Legal RAG

Anonymous Authors[1]

Abstract

Retrieval-Augmented Generation (RAG) is an effective approach for legal question answering, yet standard RAG struggles with the lexical gap between colloquial fact patterns and formal statutory corpora. While retrieving with pseudo-documents (e.g., HyDE) proves successful in general-domain RAG, it fails in legal contexts due to this lexical mismatch. To address this, we introduce SCOPE (Snap-answer COnditioned Pseudo-document Embedding), which first generates a direct "snap answer" reflecting immediate legal intuition. A pseudo-document is then generated in the style of formal legal authority to support this snap answer. Crucially, only the pseudo-document is embedded for retrieval. During the final generation phase, the snap answer and pseudo-document are discarded, ensuring the ultimate response is grounded strictly in the retrieved evidence. We evaluate SCOPE across two extremes of the question-corpus lexical gap. On BarExamQA, with low lexical similarity, SCOPE significantly bridges this gap, lifting gold passage retrieval (Hit@5) from a 1.4% baseline to 9.5%–12.1% across 8B, 26B, and 70B parameter models, while consistently improving final answer accuracy. These results demonstrate that SCOPE is a robust and tailored solution for bridging the colloquial-to-statutory vocabulary gap in complex Legal RAG challenges.

1. Introduction

Retrieval-Augmented Generation (RAG) has emerged as a powerful paradigm for enhancing Large Language Model (LLM) generation by incorporating external evidence retrieved during inference. RAG is particularly well-suited for legal question answering, as responses frequently depend on statutes, regulations, court holdings, and doctrinal rules. This domain-specific information typically falls outside the core knowledge base of LLMs, particularly reasoning models. A standard RAG pipeline typically generates an initial query to retrieve relevant external evidence, which is then concatenated into the LLM's context (Lewis et al., 2020). However, formulating an effective initial query is often the most challenging step. For example, a bar-exam fact pattern might emphasize concrete narrative details such as a driver, crutches, and a banana peel, whereas the necessary supporting passage articulates an abstract, well-established rule regarding proximate cause. Therefore, the standard RAG is prone to generating inappropriate or superficial queries for complex legal facts, leading to retrieval failures.

A common solution is to let the model first *write* the kind of passage it expects to find and search with this text instead of the raw question. HyDE (Hypothetical Document Embeddings) prompts the model to write a passage to answer the question, and uses its embedding to retrieve actual passages (Gao et al., 2023). Similarly, query expansion rewrites a pseudo-document as the query for search (Wang et al., 2023). While improving retrieval accuracy in general domains, they present significant challenges in legal contexts, where colloquial fact patterns should be mapped to formal statutory language. In specific legal cases, the reformulation step frequently breaks down. Consequently, they struggle to find correct statutory expressions, often abstracting away critical anchors, applying the wrong legal frame, or retrieving plausible but irrelevant authority.

To address that issue, we introduce SCOPE (Snap-answer Conditioned Pseudo-document Embedding), a decoupled, two-stage query generation method designed to simulate expert legal reasoning. When facing a complex fact pattern, a legal practitioner typically forms a quick initial intuition, uses that intuition to locate controlling authority, and subsequently evaluates the actual evidence on its own merits rather than anchoring on their first guess. SCOPE structurally encodes this cognitive loop. In the first phase, the model is prompted to generate a "snap answer", a rapid, unverified draft response reflecting its immediate legal intuition based purely on internal knowledge. Conditioned directly on this initial snap answer, the model then generates a pseudo-document written in the formal, abstract style of the supportive legal authority that would resolve

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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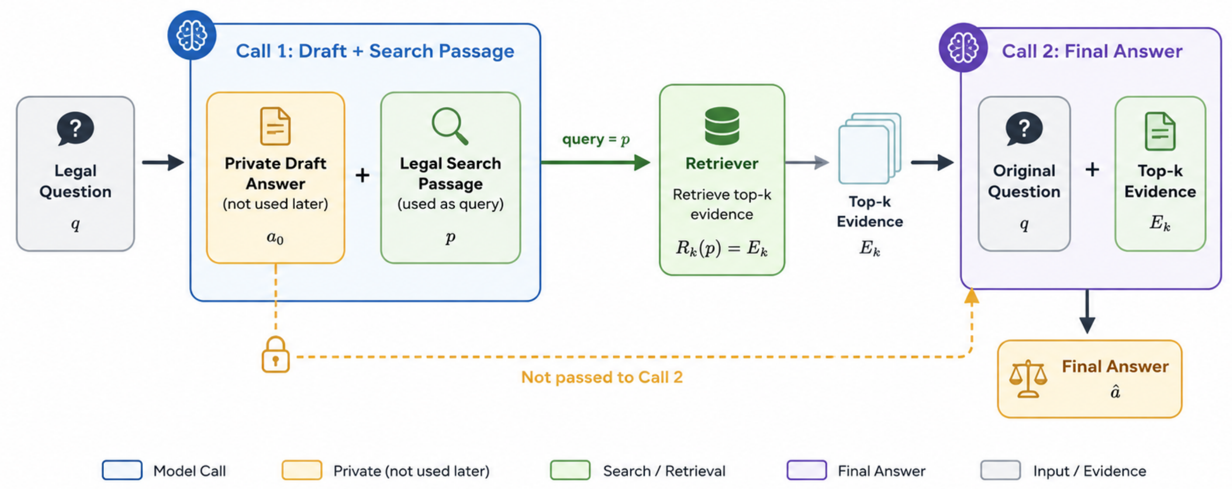

Figure 1. SCOPE: the first model call writes a private draft answer and a separate legal search passage; only the search passage is used to retrieve evidence. The second model call sees only the original question and the top-k retrieved passages, and produces the final answer. The draft is never passed to the second call.

the case. Crucially, during the retrieval phase, the snap answer is discarded to prevent confirmation bias. The pseudo-document is embedded to retrieve actual legal evidence from the corpus. In the final answer phase, a separate model call generates the final answer conditioned on the retrieved external evidence and the original question, ensuring the final reasoning is grounded in real authority and remains uncontaminated from the noisy initial guess.

We evaluate two benchmarks selected to represent opposite ends of the question-corpus vocabulary gap (Zheng et al., 2025). The first benchmark, BarExamQA, exemplifies the *weak-query regime*: its questions consist of narrative fact patterns detailing accidents, transactions, and procedures in everyday prose, whereas the target corpus relies on formal doctrinal vocabulary (e.g., “proximate cause,” “hearsay exception,” or “adverse possession”). Consequently, the raw question and the relevant legal rule share extremely low lexical similarity. In contrast, HousingQA represents the *strong-query regime*. These statutory-entailment questions inquire directly about specific legal conditions (e.g., “Can a landlord evict a tenant for committing waste?”). Because the queries explicitly utilize formal terminology, their vocabulary naturally overlaps with the target housing statutes required to answer them.

On the BarExamQA benchmark, where questions show a large gap with the corpus, SCOPE yields substantial improvements. Across the 8B, 26B, and 70B model sizes, it lifts the gold passage Hit@5 rate from a baseline of 1.4% to 9.5%-12.1%, consistently improving final answer accuracy at every model scale.

In summary, we formalize SCOPE as a decoupled, two-stage RAG framework, which is highly tailored for Legal RAG challenges. Crucially, when user questions describe scenarios using narrative prose that lacks lexical overlap with formal legal texts, SCOPE successfully bridges this gap to drive major improvements in gold passage retrieval, subsequently boosting final answer accuracy.

2. Background

Retrieval-augmented generation (RAG) couples a parametric generator with a searchable evidence store (Lewis et al., 2020). In raw-question RAG, the question q is also the retrieval query:

$$E_k^{\text{raw}} = R_k(q), \qquad \hat{a}^{\text{raw}} = G_\theta^{\text{answer}}(q, E_k^{\text{raw}}). \quad (1)$$

The system asks the corpus with the user’s words, then asks the model to answer from the top-k passages.

This interface fits law because many answers turn on identifiable authority: a statute, regulation, holding, rule statement, or doctrinal test. Its weakness is query formation. A legal question may describe concrete facts in ordinary language, while the corpus states the answer in formal legal terms. A torts fact pattern can mention crutches, a fall, and a broken leg, while the relevant support passage uses language such as proximate cause and foreseeable intervening injury. In that setting, the raw question can be a poor retrieval query even when the answer is well defined by the corpus.

Generated-query methods address this vocabulary gap by asking the model to write text that should retrieve useful evidence. HyDE generates a hypothetical document p_{hyde} and embeds that text for retrieval (Gao et al., 2023):

$$p_{\text{hyde}} = G_\theta^{\text{hyde}}(q), \qquad E_k^{\text{hyde}} = R_k(p_{\text{hyde}}). \tag{2}$$

Query2doc and related expansion methods use a similar idea, turning sparse query text into a richer retrieval handle (Wang et al., 2023). For legal tasks, the useful generated text is usually not a longer version of the question. It is closer to the style of the authority the retriever should find.

SCOPE belongs to this generated-query family, but changes the first call. The model first forms a private snap answer and then writes a separate legal search passage conditioned on that tentative legal frame. The draft answer is not evidence and is withheld from the final answer call. The comparisons in this paper therefore separate four roles: LLM-only tests the model without retrieval, raw-question RAG tests whether the question is already a good corpus query, HyDE tests generated legal prose without the snap-answer signal, and Gold Evidence supplies the labeled support passage directly as an evidence-availability control.

3. SCOPE

3.1. Design Rationale

SCOPE mirrors a small piece of how an expert reads a legal scenario. A practitioner facing a fact pattern forms a quick intuition about what is at issue (e.g. proximate cause, a hearsay exception, or a state-specific notice rule), looks up the controlling authority by that intuition, and then judges the evidence without allowing intuition to bias the judgment. SCOPE encodes the same loop. The first call asks the model to form an initial draft answer from its priors and to write a separate legal search passage for retrieval. The second call reasons only over the question and the retrieved evidence; it never sees the draft answer, so the final answer is judged against what the corpus actually says. Figure 1 shows the resulting two-call pipeline.

3.2. Two Calls, One Retrieval Query

SCOPE separates query formation from final answering. The same model G_θ is used for both calls; only the prompt template differs (the superscript). Given a formatted legal question q, the first call produces a private draft answer a_0 and a legal search passage p:

Algorithm 1 SCOPE pipeline for one legal question.

Require: question q; generator G_θ; encoder enc; corpus $\mathcal{C}$; depth k

1: $(a_0, p) \leftarrow G_\theta^{\text{snap}}(q)$ // first call
2: **validate** a_0 has a parseable answer label and p contains no answer marker
3: $E_k \leftarrow \text{TopK}_{c\in\mathcal{C}}\langle \text{enc}(p), \text{enc}(c)\rangle$ // retrieve using p only
4: **rerank** E_k by cross-encoder on p
5: $\hat{a} \leftarrow G_\theta^{\text{answer}}(q, E_k)$ // second call; a_0 is discarded
6: **return** $\hat{a}$

$$(a_0, p) = G_\theta^{\text{snap}}(q). \tag{3}$$

Only p is exposed to retrieval. Let $\text{enc} : \mathcal{T} \to \mathbb{R}^d$ denote the dense encoder shared by queries and corpus passages, and let $\mathcal{C}$ denote the searchable corpus. The retriever ranks every $c \in \mathcal{C}$ by similarity to the embedded search passage and returns the top-k candidates,

$$E_k = \operatorname*{TopK}_{c\in\mathcal{C}} \big\langle \text{enc}(p), \text{enc}(c)\big\rangle, \tag{4}$$

after which a cross-encoder reranker reorders E_k on the same passage p. The final answer is then

$$\hat{a} = G_\theta^{\text{answer}}(q, E_k), \tag{5}$$

with a_0 withheld from the final context. The first call can use the model's prior knowledge of law to choose vocabulary; the second call must answer from the retrieved corpus text. Algorithm 1 summarizes the pipeline.

Contrast with HyDE. HyDE writes a single hypothetical document directly from the question,

$$p_{\text{hyde}} = G_\theta^{\text{hyde}}(q), \qquad E_k^{\text{hyde}} = R_k(p_{\text{hyde}}), \tag{6}$$

and uses p_{hyde} as the retrieval handle (Gao et al., 2023). SCOPE keeps that generated-query premise but reshapes the first call: the model first commits to a tentative legal frame a_0, then writes p as language consistent with that frame. The snap draft acts as an information bottleneck that selects *what kind* of authority the passage should look like, but it is discarded before the corpus is queried and never reaches the answer call. For a torts fact pattern about a broken leg, crutches, and a later fall, this turns an accident-heavy query into a passage about proximate cause and foreseeable subsequent injury (Figure 2).

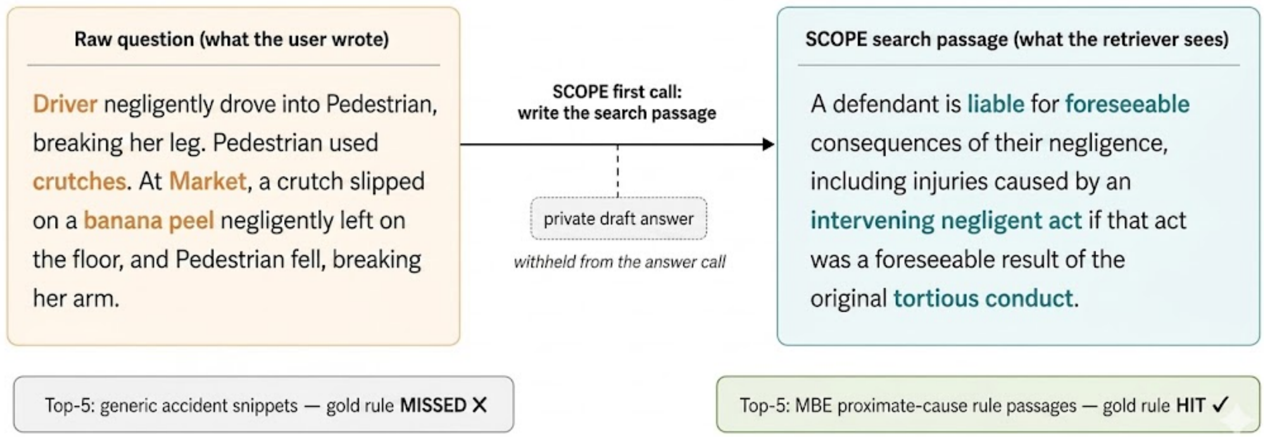

Figure 2. What the two methods send to the retriever for one BarExamQA item (`mbe_1175`). Raw-question RAG embeds the narrative fact pattern; SCOPE embeds a short doctrinal passage about foreseeable intervening injury, which is the language of the rule the retriever actually needs to surface.

3.3. Grounding Guardrails

The two-call structure gives the method two guardrails that follow from the formulation above. First, the draft a_0 is private: it can shape what to look for, but it is not handed in as evidence. The second call's answer depends on the draft only through the search passage and the retrieved set,

$$\hat{a} \perp\!\!\!\perp a_0 \;\Big|\; (q, E_k), \tag{7}$$

so the model cannot anchor its evidence-stage reasoning on a guess it made without seeing the corpus. Second, the search passage p is written like legal authority rather than chain-of-thought reasoning: it names the rule, statutory condition, or doctrinal test that would resolve the question, not the steps the model used to get there. The intuition step is allowed to use model priors and the evidence step is forced back to the corpus.

Each SCOPE response follows three steps: validate that the first call produced both a formatted draft answer and a clean search passage; retrieve and rerank using only the search passage; answer the original question from the retrieved passages without exposing the draft answer. The retrieval passage is constrained to reflect the corpus it is searching over, typically describing the controlling rule, statutory condition, legal relationship, or doctrinal test without answer labels or an `Answer:` header, as to prevent answer leakage from the first step to the second step.

4. Evaluation

4.1. Metrics

- **Hit@k.** A binary indicator of whether at least one labeled gold passage appears in the top-k retrieved results for a given query. Averaged over the evaluation set, Hit@k reports the fraction of questions for which the retriever successfully surfaces relevant evidence within the top k candidates. It measures retrieval *coverage*: whether the correct evidence is exposed to the answer model at all, but not how highly it is ranked.
- **MRR@k (Mean Reciprocal Rank).** The average of the reciprocal rank of the first relevant passage within the top-k retrieved results, where the reciprocal rank is $1/r$ if the first relevant passage appears at position $r \leq k$, and 0 otherwise. Formally,

$$\text{MRR@}k = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{r_q} \cdot \mathbb{1}[r_q \leq k], \tag{8}$$

  where Q is the set of evaluation queries and r_q is the rank of the first relevant passage for query q. MRR@k measures retrieval *ranking quality*: a gold passage ranked first contributes 1.0, while one ranked fifth contributes only 0.2. Together with Hit@k, it distinguishes retrieval that merely includes the right evidence from retrieval that promotes it to positions the answer model is more likely to attend to.

4.2. Two Legal Retrieval Regimes

BarExamQA and HousingQA sit at opposite ends of the question-corpus vocabulary gap. BarExamQA questions are real life scenarios that must be interpreted in the context of the law, and the support corpus uses doctrinal vocabulary

Table 1. Main answer accuracy (%). Bold marks the largest value in each column among the deployable methods. The Gold Evidence row–where we provide the model the gold passage–is set apart by a rule and is not eligible for bolding. Averages are descriptive over available cells.

	BarExamQA				HousingQA			
Method	8B	26B	70B	Avg.	8B	26B	70B	Avg.
LLM	**57.3**	80.8	78.7	72.3	55.4	56.1	44.8	52.1
Raw question RAG	54.5	78.0	74.6	69.0	**62.3**	**66.1**	62.1	**63.5**
HyDE	56.1	80.3	**80.2**	72.2	59.1	65.0	**62.2**	62.1
SCOPE (ours)	56.9	**82.0**	79.7	**72.9**	59.0	65.1	59.6	61.2
Gold Evidence	60.0	78.6	79.2	72.6	64.3	–	67.3	65.8

(Zheng et al., 2025). HousingQA questions are statutory-entailment questions framed in landlord-tenant terms, sharing vocabulary with the housing statutes that answer them, so raw queries are already corpus-shaped. The main paper reports both benchmarks (Zheng et al., 2025).

BarExamQA. BarExamQA contains 1,195 multiple-choice bar-exam questions in the evaluation split used here. The retrieval corpus is an unstructured collection of legal passages, with some passages labeled 'gold passages' by legal experts that are intended to be the ideal evidence for answering one of the questions (Zheng et al., 2025). Hit@5 and MRR@5 metrics therefore measure whether the retriever exposes the labeled rule passage before the answer model chooses an option.

HousingQA. HousingQA contains 6,853 yes/no statutory entailment questions over state housing statutes (Zheng et al., 2025). The corpus contains many jurisdictions, and each question names the relevant state. All main HousingQA retrieval rows apply that state as a metadata filter before ranking.

4.3. Models, Baselines, and Retrieval Stack

We evaluate three model configurations: Llama 3.1 8B Instruct (Grattafiori et al., 2024), Gemma 4 26B, and Llama 3.3 70B Versatile. Within each response, the same model serves as both the query generator (when one is needed) and the final answer model, so accuracy lift cannot be attributed to a stronger reader.

We compare four deployable retrieval methods and providing models with the gold passage. Models using retrieval methods answer from $k = 5$ retrieved passages.

- **LLM only** answers the original question without retrieval.
- **Raw question RAG** retrieves with the question text, then answers from the question plus top-k evidence.
- **HyDE** retrieves with a generated legal passage, without a draft answer prior.
- **SCOPE (ours)** retrieves with a draft-guided legal search passage; the draft answer is private and is not shown to the final answer call.
- **Gold Evidence** supplies the labeled gold passage directly, simulating 'perfect retrieval'.

Structured rewrite and gold-plus-neighbor (using gold passage to retrieve 4 more passages) diagnostics appear in the appendix; route labels are retained in the released artifacts for reproducibility.

Retrieval stack. Runs use seed 42, temperature 0, and a 2048-token answer cap. If the model produces a malformed generated passage or no parseable final answer, the example is retried. Retrieval uses Chroma collections built with `Alibaba-NLP/gte-large-en-v1.5` (Zhang et al., 2024; Li et al., 2023) embeddings; a cross-encoder (`ms-marco-MiniLM-L-6-v2`) then reranks the candidate set before the top-k cut.

Generated-query evaluation. Generated-query methods are evaluated in two stages. First, the query-generation call writes the HyDE passage or SCOPE search passage. Then retrieval stores the ranked document ids. The answer run replays those ids, and that evidence is used in final question answering.

Per-dataset retrieval interfaces. BarExamQA retrieval uses the shared fact pattern and question stem; generated methods may see option text but no correctness marker. HousingQA retrieval applies a state metadata filter. The filter is not a SCOPE-specific feature; it is applied to every HousingQA row to prevent retrieval of evidence from a different jurisdiction.

4.4. Answer and Evidence Metrics

The downstream metric is exact answer accuracy. BarExamQA uses the last valid multiple-choice answer line, and HousingQA uses `Answer:Yes` or `Answer:No`. Retrieval is measured by Hit@5 and MRR@5: whether any labeled support passage appears in the top five, and how early it appears.

We also report efficiency as the number of model calls per question, the tokens the answer call processes per question (the "answer-stage tokens"), and correct answers per million answer-stage tokens. Two-call methods are not penalized in this count: the first-stage query-generation call is counted as part of the method footprint and logged separately. The answer-stage token count shows how much context the final answer call sees once retrieval has selected evidence.

The main tables include only validated full complete conditions. Appendix A.1 lists the inclusion rule, coverage grid, supplemental controls, top-k diagnostics, efficiency table, and reproducibility details.

5. Results

5.1. When SCOPE Helps Answers

SCOPE helps in proportion to the question-corpus vocabulary gap. On BarExamQA, fact patterns describe accidents and transactions in narrative prose, while the support corpus uses doctrinal language; raw-question RAG retrieves the labeled rule passage in only 1.4% of cases at Hit@5. SCOPE raises Hit@5 to 9.5%, 12.1%, and 11.0% across the 8B, 26B, and 70B models, and converts those retrieval gains into answer gains of +2.4pp, +4.0pp, and +5.1pp over raw-question RAG; the Gemma 4 26B improvement is paired-significant at McNemar $p < 0.001$. The lift grows with model size: larger generators both write more useful search text and use the retrieved evidence more effectively.

HousingQA sits at the opposite end of that axis. Statutory-entailment questions already share vocabulary with the target housing statutes; raw-question retrieval reaches 36.9% Hit@5 with no query generation. With less of a gap to close, SCOPE delivers answer parity rather than lift. §5.6 shows that even the small HousingQA gap can be closed: a single statute-style corpus example in the SCOPE system prompt recovers Hit@5 above the raw retrieval baseline.

Table 2 makes the lexical gap concrete on a single BarExamQA item: the fact pattern speaks of crutches and a banana peel, while the controlling rule speaks of proximate cause and foreseeable intervening injury. Raw retrieval pulls generic accident material, and HyDE drifts toward proximate cause without landing on the rule; both pick the wrong option. SCOPE writes a search passage in MBE proximate-cause language, recovers the gold rule, and flips the answer to correct.

5.2. Did the Right Evidence Appear?

Table 4 separates retrieval from answer selection. Figure 3 extends that view across k from one to ten for both datasets. On BarExamQA, generated legal passages move every top-k metric up: SCOPE and HyDE both raise Hit@5 by roughly an order of magnitude over raw retrieval, with SCOPE slightly ahead at every depth. On HousingQA the picture is reversed because raw queries already share vocabulary with the corpus, and the generated-query rows trade some labeled-evidence exposure for broader corpus context that the answer model still uses well (parity in Table 1).

SCOPE converts a +9.5pp Hit@5 and +4.8pp MRR@5 retrieval shift into a +3.9pp average BarExamQA answer accuracy gain over raw-question RAG, while exact gold evidence still appears in only about one in nine top-five lists. The remaining gap points to ranking, option mapping, and evidence use.

Table 2. Worked BarExamQA example (`mbe_1175`, Gemma 4 26B). Gold answer: `B` (driver liable for both injuries).

Method	Pred.	Top-5 retrieval
Raw question RAG	`D` ✗	Generic accident and intervening-cause material; gold rule passage missed.
HyDE	`D` ✗	Drifts toward proximate cause but stays broad; gold rule passage missed.
SCOPE (ours)	`B` ✓	Anchored on foreseeable subsequent injury; surfaces MBE proximate-cause rule passages.

Table 3. Answer-stage token accounting for rows with token logs. First-stage query-generation usage is excluded. All methods use one scored answer call per question except Rewrite, which uses two. Bold marks the highest *Correct / 1M tok.* among single-call non-gold evidence retrieval methods (Raw question RAG, HyDE, SCOPE).

Method	Cells	Input tok./q	Output tok./q	Correct / 1M tok.
LLM	5	131	237	1487.2
Raw question RAG	5	2295	339	241.5
HyDE	5	2237	340	244.5
SCOPE (ours)	4	2001	376	**268.3**
Gold Evidence	5	1447	215	404.5

5.3. Differentiation from Previous Methods

On this item, raw-question RAG retrieves generic accident material and HyDE drifts toward proximate cause without landing on the rule. SCOPE writes a short passage about foreseeable intervening injury, surfaces the MBE proximate-cause rule, and picks (B). The win comes from a more directed search query than HyDE or the raw question produces.

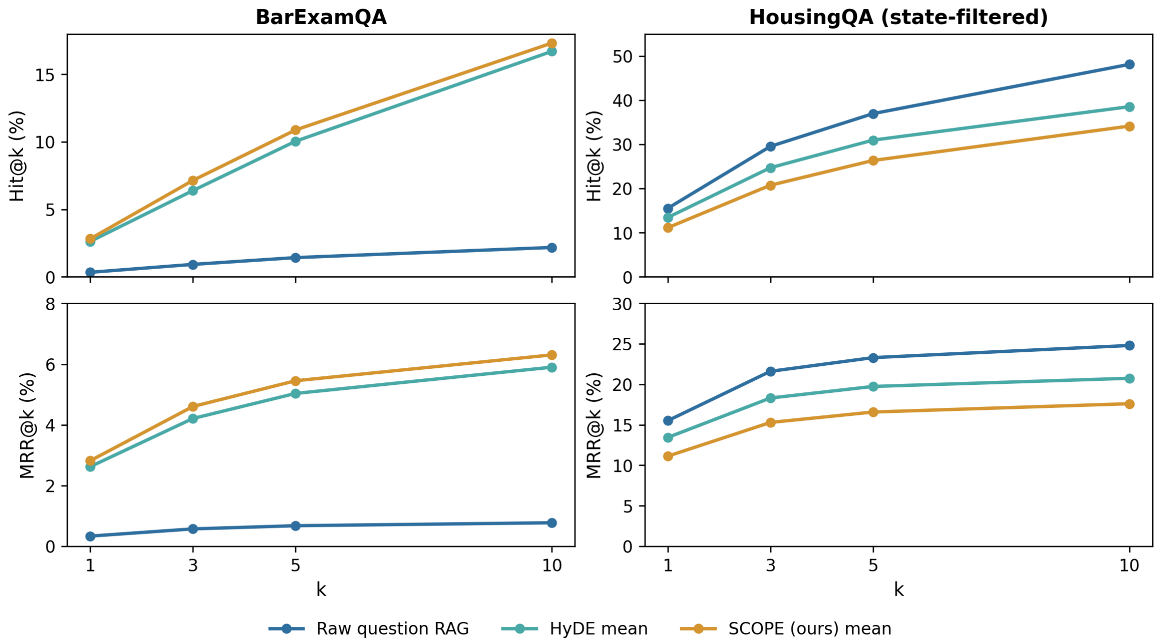

Figure 3. Top-k retrieval exposure and rank quality for full top-10 caches. BarExamQA generated-query curves average over Llama 3.1 8B, Gemma 4 26B, and Llama 3.3 70B; HousingQA averages over Llama 3.1 8B, Gemma 4 26B, and Llama 3.3 70B with the jurisdiction state filter.

Table 4. Gemma 26B evidence exposure at $k = 5$ (%). Bold marks the largest value in each column. HousingQA uses jurisdiction state filtering for all retrieval methods.

	BarExamQA		HousingQA	
Method	Hit@5	MRR@5	Hit@5	MRR@5
Raw question RAG	1.4	0.7	36.9	23.3
HyDE	11.4	5.4	30.6	19.6
SCOPE (ours)	**12.1**	**6.0**	**38.1**	**24.5**

5.4. SCOPE Matches the Gold Evidence on Larger Models

On BarExamQA, SCOPE matches or exceeds the labeled gold passage on the two larger models: 82.0% versus 78.6% on Gemma 4 26B, and 79.7% versus 79.2% on Llama 3.3 70B. Only Llama 3.1 8B trails the gold evidence method. The retrieved top-5 list does more work than the labeled rule alone: surrounding legal passages supply context the answer model uses. The same pattern shows up at the gold evidence level in Appendix A.2: adding neighbors to the gold passage also improves accuracy on Llama 3.1 8B and Gemma 4 26B.

5.5. Leading Answer-Token Efficiency

Table 3 reports correct answers per million tokens that the answer call processes. Among non-gold evidence methods SCOPE is the most token-efficient at 268.3 correct per million answer-stage tokens, versus 244.5 for HyDE and 241.5 for raw-question RAG: the reranked short passages produce a shorter average answer context at higher accuracy. The gold evidence method reaches 404.5 correct per million tokens because it sees only the single labeled passage.

5.6. One-Shot Corpus Inclusion

Table 5. Statute/passage exemplar lift, Gemma 4 26B N = 500 retrieval probes (Hit@5 / MRR@5, %). Δ is the exemplar lift over canonical SCOPE. Bold marks the higher of SCOPE vs. +Exemplar within each metric per row. Reference raw-question Hit@5: BarExamQA 1.6, HousingQA (state-filtered) 36.9, Legal-Link-EU 90.0.

Dataset	SCOPE	+ Exemplar	Δ Hit@5 / MRR@5
BarExamQA	13.0 / **6.4**	**13.6** / 6.3	+0.6 / −0.1
HousingQA	38.2 / 24.3	**41.2** / **26.5**	+3.0 / +2.2
Legal-Link-EU	68.2 / 55.6	**75.8** / **62.6**	+7.6 / +7.0

The first call's prompt can be improved with a single example. We add one real passage from the target corpus to the SCOPE generation prompt (with its question, answer, and document id stripped) and re-run the first call. On 500-question probes for Gemma 4 26B, this lifts retrieval on every benchmark we tested (Table 5); the size of the lift tracks how unfamiliar the target corpus phrasing is. Paired answer accuracy on the HousingQA 500-question slice is 62.8% with the example versus 63.0% for plain SCOPE (-0.20pp): the example recovers retrieval without costing answer accuracy. On an Ohio writ-of-execution question, for example, the prompted passage names "writ of restitution" and "execution" in statute style and retrieves the gold Ohio statute at rank 3, where plain SCOPE drifts to neighboring writ-of-possession language.

Legal-Link-EU, another benchmark briefly tested, also shows SCOPE's limitations. Its questions cite source and target document anchors directly, so the raw question is already a strong corpus search (Hit@5 90.0% versus plain SCOPE 68.2%). When the user has already framed their need as a search query, the intuition step is not needed. The typical legal scenario in practice is the other way around: a fact pattern, a tenant complaint, or an audit notice that the user has to turn into a search query before any retrieval can happen. That is where SCOPE targets its gains.

6. Related Work

RAG and generated queries. RAG conditions generation on retrieved nonparametric memory (Lewis et al., 2020), while reader architectures such as Fusion-in-Decoder show that answer quality depends on how the generator uses retrieved passages (Izacard & Grave, 2021). HyDE improves dense retrieval by generating a hypothetical document and searching with that text (Gao et al., 2023); Query2doc uses related query-expansion ideas (Wang et al., 2023). SCOPE keeps the generated-query premise but conditions the retrieval passage on a private preliminary answer, then removes that answer before final generation.

RAG control and correction. Several systems add control decisions around retrieval. FLARE (Jiang et al., 2023a) retrieves during generation when upcoming content is uncertain (Jiang et al., 2023b); Self-RAG learns reflection signals for retrieval and critique (Asai et al., 2024); Corrective RAG evaluates retrieved documents before continuing generation (Yan et al., 2024). SCOPE instead uses a fixed two-call pipeline so the experiment isolates one question: whether snap-conditioned legal search text is a better retrieval handle.

Legal grounding benchmarks. LegalBench, LexGLUE, and Pile of Law standardized legal reasoning, classification, and corpus-grounded language tasks (Guha et al., 2023; Chalkidis et al., 2022; Henderson et al., 2022). Work on legal assistant reliability shows why source grounding remains necessary (Magesh et al., 2024). LegalBench-RAG and Legal RAG Bench argue that legal RAG should be evaluated on retrieval and answer quality together (Pipitone & Alami, 2024; Butler & Butler, 2026). BarExamQA and HousingQA make the retrieval interface itself a central object of study: the question may or may not already look like the authority it needs to find (Zheng et al., 2025).

7. Conclusion

SCOPE encodes a small piece of expert legal reasoning into a fixed two-call pipeline: form an intuition about the controlling doctrine, retrieve evidence by that frame, then judge the evidence on its own without anchoring to the intuition. The lift over raw-question RAG tracks the question-corpus vocabulary gap, because that gap is precisely the regime in which the intuition step matters. When the raw question is a poor retrieval handle, letting the model form an intuition before it searches and forcing the final answer back to the corpus delivers both retrieval and answer gains; when the question already reads like a passage from the corpus, SCOPE reaches parity with raw-question RAG and the one-example variant from Section 5.6 recovers retrieval.

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A. Appendix

A.1. Result Inclusion Policy

A row is used as a paper claim only when the result record names the dataset, provider, method, answer depth, metric values, and any row note needed to interpret the result. Rows without validated full results are omitted from the main comparison rather than imputed.

- Main benchmarks: BarExamQA and HousingQA.
- Route labels: `groq-llama8b`, `or-gemma4-26b`, `groq-llama70b`.
- Main modes: `llm_only`, `rag_simple`, `rag_hyde`, `SCOPE`, and `golden_passage`. `rag_rewrite` is a supplemental generated-query control, and `golden_plus_neighbors` is retained only as a neighbor-dilution diagnostic.
- Answer depth: $k = 5$ for retrieval rows.
- HousingQA interface: retrieval rows must use the jurisdiction state filter to count as main-interface rows.

The main matrix has 35 signed cells out of 42 expected (BarExamQA 21/21; HousingQA 14/21).

A.2. Supplemental Controls

The supplemental tables separate the fixed-method comparison from controls that are useful for interpreting failure modes: structured rewrites, gold evidence context, and neighbor dilution.

Table 6. Dataset-level and pooled deltas. Mean percentage-point differences across signed model slices.

Slice	Snap vs. raw pp	HyDE vs. Snap pp	Gold vs. raw pp
BarExamQA	+3.8 (n=3)	-0.7 (n=3)	+3.6 (n=3)
HousingQA	-2.9 (n=2)	+1.4 (n=2)	+3.6 (n=2)
Pooled	**+1.1 (n=5)**	**+0.1 (n=5)**	**+3.6 (n=5)**

Table 7. SCOPE against non-gold evidence controls. Cells are accuracy percentages. Bold marks the strongest result among the main deployable methods for each dataset–model pair.

	BarExamQA			HousingQA	
Method	Llama 3.1 8B	Gemma 4 26B	Llama 3.3 70B	Llama 3.1 8B	Llama 3.3 70B
LLM	**57.3**	80.8	78.7	55.4	44.8
Raw question RAG	54.5	78.0	74.6	**62.3**	62.1
HyDE	56.1	80.3	**80.2**	59.1	**62.2**
Rewrite	57.3	80.7	77.2	–	–
SCOPE (ours)	56.9	**82.0**	79.7	59.0	59.6
Snap vs. raw (pp)	+2.4	+4.0	+5.1	-3.3	-2.5

Table 8. Gold evidence controls. Accuracy values are percentages; deltas are percentage points. Bold marks the highest accuracy in each column across Raw question RAG, SCOPE, and Gold.

	BarExamQA			HousingQA	
Method	Llama 3.1 8B	Gemma 4 26B	Llama 3.3 70B	Llama 3.1 8B	Llama 3.3 70B
Raw question RAG	54.5	78.0	74.6	62.3	62.1
Gold Evidence	**60.0**	78.6	79.2	**64.3**	**67.3**
SCOPE (ours)	56.9	**82.0**	**79.7**	59.0	59.6
Gold vs. raw (pp)	+5.5	+0.6	+4.6	+2.0	+5.2

Table 9. Neighbor-dilution diagnostic on BarExamQA. Positive deltas indicate that retrieving neighboring passages using the gold passage as the query improves over providing only the gold passage. Bold marks the higher accuracy for each model.

Method	Llama 3.1 8B	Gemma 4 26B	Llama 3.3 70B
Gold	60.0	78.6	**79.2**
Gold + neighbors	**61.8**	**80.7**	77.8
Δ (pp)	+1.8	+2.1	-1.4

A.3. Top-k Retrieval Diagnostics

Tables 10 and 11 give the underlying per-row retrieval exposure as the cutoff grows from one to ten passages; the same data drives Figure 3 in Section 5.

Table 10. HousingQA raw retrieval with and without the jurisdiction filter. Values are percentages from complete top-10 caches. Bold marks the higher value in each column.

Corpus scope	Hit@1	Hit@3	Hit@5	Hit@10	MRR@10
National corpus	0.8	1.9	2.8	5.1	1.8
Jurisdiction metadata filter	**15.5**	**29.5**	**36.9**	**48.1**	**24.8**

BarExamQA generated passages raise the chance of seeing the labeled evidence throughout the top-10 list, although MRR remains modest because the exact supporting passage is rarely ranked first. HousingQA behaves differently: the jurisdiction filter is the major retrieval correction, and raw state-filtered questions already retrieve relevant evidence often.

Table 11. Top-k retrieval diagnostics from complete top-10 caches. Values are percentages. For generated-query methods, "Mean" averages model-specific retrieval caches. Bold marks the highest value within each comparable scope.

Model / Scope	Method	Hit@3	Hit@5	Hit@10	MRR@10
BarExamQA					$n = 1195$
Shared / Mean over 3 models	Raw question RAG	0.9	1.4	2.2	0.8
	HyDE	6.4	10.0	16.7	5.9
	SCOPE (ours)	**7.1**	**10.9**	**17.3**	**6.3**
Llama 3.1 8B	HyDE	5.5	8.3	13.5	5.2
	SCOPE (ours)	**6.3**	**9.5**	**14.8**	**5.4**
Gemma 4 26B	HyDE	7.1	11.4	**19.1**	6.4
	SCOPE (ours)	**7.9**	**12.1**	18.7	**6.9**
Llama 3.3 70B	HyDE	6.5	10.5	17.6	6.1
	SCOPE (ours)	**7.2**	**11.0**	**18.5**	**6.6**

A.4. Statute/Passage Exemplar Probe Details

The exemplar variant adds one fixed sanitized real-passage exemplar to the SCOPE generation prompt, with the exemplar's own question, answer, and document id stripped. The same q500 retrieval interfaces are used as the corresponding canonical SCOPE rows: BarExamQA uses the shared support-passage collection; HousingQA uses the jurisdiction state filter; Legal-Link-EU uses the EUR-Lex evidence corpus with the same `CROSS_ENCODER_MAX_CHARS=22000` setting as the main SCOPE cache. The retrieval probes are computed directly from the cached passage-id lists; paired answer accuracy was scored only on the HousingQA q500 slice (62.8% exemplar vs 63.0% canonical SCOPE, $-0.20\mathrm{pp}$, $p = 1.0$). Table 5 in Section 5.6 reports the lift values.

A.5. Reproducibility Statement

All tables and figures are derived from the result records used for this draft. The released package will include the records, plotting code, and paper sources needed to reproduce the reported matrices and figures.